MIT Coding Challenge: Monetary Policy Shocks and Exchange Rates

Complete Output Compilation

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Contents

Part I
Tables

Table 1: Summary Statistics: Monetary Policy Surprises and Asset Price Changes on FOMC Days (1994–2026)

	N	Mean	Std	Min	P25	Median	P75	Max
STMT (bps)	274	-0.000	3.679	-26.444	-1.038	0.463	1.586	9.010
MP1 (bps)	274	-1.044	6.795	-46.500	-0.840	0.000	0.500	16.333
MP2 (bps)	274	-0.785	4.712	-38.984	-1.000	0.000	0.814	15.100
$\mathrm{d}_C AD$	270	-0.061	0.524	-5.072	-0.269	-0.028	0.135	1.836
$\mathrm{d}_J PY$	270	0.015	0.585	-2.544	-0.273	0.058	0.352	1.864
$\mathrm{d}_M XN$	270	-0.027	1.221	-5.474	-0.428	-0.101	0.186	13.066
$\mathrm{d}_N OK$	270	-0.089	0.688	-3.376	-0.384	-0.030	0.327	1.554
$\mathrm{d}_C HF$	270	-0.059	0.559	-3.229	-0.311	0.010	0.250	1.253
$\mathrm{d}_A UD$	270	-0.046	0.906	-6.667	-0.429	-0.088	0.259	7.569
$\mathrm{d}_E UR$	229	-0.089	0.467	-2.961	-0.238	-0.043	0.130	1.245
$\mathrm{d}_G BP$	270	-0.071	0.548	-4.435	-0.346	-0.048	0.208	2.161
$\mathrm{d}_U ST_2 Y$	272	-0.816	7.450	-28.000	-4.250	0.000	3.000	27.000
$\mathrm{d}_U ST_5 Y$	272	-0.974	8.342	-46.000	-6.000	0.000	3.000	25.000
$\mathrm{d}_U ST_1 0Y$	272	-0.710	7.309	-51.000	-4.000	0.000	3.000	22.000
$\mathrm{d}_B E_5 Y$	194	0.211	4.897	-20.000	-2.000	1.000	2.000	29.000
$\mathrm{d}_B E_1 0Y$	194	0.237	3.719	-15.000	-2.000	0.000	2.000	11.000

1 Table 1: Summary Statistics

2 Table 2: Regression Results

Table 2: Asset Price Responses to Monetary Policy Surprises

	$\hat{\beta}$	Robust SE	t-stat	N	R^2
Panel A: Treasury Yields (bps)					
UST 2Y	1.058***	(0.172)	6.15	272	0.270
UST 5Y	0.903***	(0.177)	5.11	272	0.157
UST 10Y	0.507***	(0.146)	3.47	272	0.065
Panel B: Breakeven Inflation (bps)					
BE 5Y	-0.420	(0.262)	-1.60	194	0.060
BE 10Y	-0.260**	(0.108)	-2.41	194	0.040
Panel C: Exchange Rates (% spot return, $r = -\Delta\log(e)$)					
EUR	−0.0056	(0.0081)	−0.69	229	0.002
GBP	−0.0025	(0.0080)	−0.31	270	0.000
JPY	−0.0162	(0.0104)	−1.55	270	0.010
CHF	−0.0153	(0.0107)	−1.43	270	0.010
AUD	0.0004	(0.0172)	0.03	270	0.000
CAD	−0.0081	(0.0119)	−0.68	270	0.003
MXN	0.0081	(0.0177)	0.45	270	0.001
NOK	−0.0051	(0.0107)	−0.48	270	0.001

Notes: Robust (HC1) standard errors in parentheses. STMT is in basis points; β represents the asset response to a 1 bp statement surprise. FX returns use spot return convention ($r = -\Delta\log(e)$, positive = foreign appreciation); negative β indicates foreign depreciation on hawkish surprise. Sample: FOMC announcement days, 1994–2024. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

3 Table 2a: NFA by Country

Table 3: Net Foreign Asset Positions by Country

Country	Currency	Avg NFA/GDP (%)
Switzerland	CHF	+142.3
Norway	NOK	+67.8
Japan	JPY	+58.2
Canada	CAD	+12.4
United Kingdom	GBP	−8.5
Eurozone	EUR	−12.3
Mexico	MXN	−38.6
Australia	AUD	−52.1
Cross-country mean		+8.9
Cross-country std dev		69.8

Notes: NFA/GDP from External Wealth of Nations database (Lane & Milesi-Ferretti 2018), averaged over sample period 1994–2024. Positive values indicate net creditor positions; negative values indicate net debtor positions. Sample includes G10 currencies excluding SEK and NZD due to data availability.

4 Table 2b: Treasury Response

Table 4: Treasury Yield Responses to Monetary Policy Surprises

	STMT		MP1	
Maturity	β	R^2	β	R^2
2-year	1.058***	0.270	0.478***	0.163
5-year	0.903***	0.157	0.412***	0.152
10-year	0.507***	0.065	0.287***	0.112

Notes: OLS regression of Treasury yield change (basis points) on monetary surprise (basis points) on FOMC announcement days only. Heteroskedasticity-robust (HC1) standard errors. STMT is the statement/PC surprise; MP1 is the target rate surprise. The declining response with maturity is consistent with monetary surprises primarily shifting near-horizon expected policy rates. Sample: 272 FOMC announcements, 1994–2024. *** $p < 0.01$.

5 Table 2c: FX Response

Table 5: Exchange Rate Responses to Statement Surprises (STMT)

Currency	β	SE	t-stat	p-value	R^2
AUD	0.0004	0.0172	0.03	0.979	0.000
CAD	−0.0081	0.0119	−0.68	0.494	0.003
CHF	−0.0153	0.0107	−1.43	0.155	0.010
EUR	−0.0056	0.0081	−0.69	0.492	0.002
GBP	−0.0025	0.0080	−0.31	0.758	0.000
JPY	−0.0162	0.0104	−1.55	0.121	0.010
MXN	0.0081	0.0177	0.45	0.650	0.001
NOK	−0.0051	0.0107	−0.48	0.634	0.001

Notes: Dependent variable is the spot return $r_{i,t} = -\Delta \log(e_{i,t})$, where $e_{i,t}$ is foreign currency per USD. Positive r indicates foreign appreciation (USD depreciation). Signs are flipped from raw log-change to match Antolín-Díaz et al. (2023) convention. Hawkish surprises (STMT > 0) $\rightarrow$ foreign depreciation ($r < 0$), so most $\beta < 0$. Heteroskedasticity-robust (HC1) standard errors. STMT is in basis points; β represents the percentage FX response to a 1 bp statement surprise. Exchange rate responses are small and imprecisely estimated in daily data ($R^2 < 1\%$ for all currencies). Sample: 269–270 FOMC events, 1994–2024.

6 Table 3: NFA Panel Regression

Table 6: Spot Returns and Net Foreign Assets

	(1) Main (Date Cl.)	(2) Two-Way (Two-Way Cl.)	(3) Country (Country Cl.)	(4) MP1 (Date Cl.)
STMT	-0.5978	-0.5978	-0.5978***	
	(0.6176)	(0.4953)	(0.2160)	
MP1				0.0216
				(0.4489)
Surprise × NFA/GDP	-0.0048	-0.0048	-0.0048	-0.0162*
	(0.0067)	(0.0036)	(0.0051)	(0.0091)
NFA/GDP	0.0004	0.0004**	0.0004***	0.0003
	(0.0004)	(0.0002)	(0.0001)	(0.0004)
Country FE	Yes	Yes	Yes	Yes
Date Cluster	Yes	Yes	No	Yes
Country Cluster	No	Yes	Yes	No
Observations	2103	2103	2103	2103
R^2	0.0014	0.0014	0.0014	0.0059

Notes: I define $e_{i,t}$ as foreign currency units per USD (e.g., JPY/USD). The dependent variable is the spot return $r_{i,t} = -\Delta \log(e_{i,t})$, so $r > 0$ indicates foreign appreciation (USD depreciation). This convention matches Antolín-Díaz et al. (2023), so that $\beta_2 > 0$ means creditor countries depreciate less. STMT is the statement surprise; MP1 is the target rate surprise, both in basis points. NFA/GDP is from Lane & Milesi-Ferretti (EWN, 2024 update), lagged one year and demeaned so β_1 represents the effect at average NFA. **Standard errors:** Column (1) clusters by FOMC date, the preferred specification because STMT_t is identical across currencies on each date, inducing cross-sectional correlation Petersen2009. Column (2) clusters two-way (country + date) as a conservative robustness check. Column (3) clusters by country only; while reported for completeness, it is inappropriate as it ignores the common-shock structure. Sample: 8 currencies (AUD, CAD, CHF, EUR, GBP, JPY, MXN, NOK) × 269 FOMC events, 1994–2024. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

7 Table 4: Time Variation (Post-GFC)

Table 7: Time Variation in the Balance-Sheet Channel

	(1) Baseline (Date Cl.)	(2) STMT + Post-GFC (Date Cl.)	(3) MP1 + Post-GFC (Date Cl.)
STMT	-0.5978	-0.3937	
	(0.6176)	(0.5155)	
MP1			0.1492
			(0.3954)
Surprise × NFA/GDP	-0.0048	-0.0152*	-0.0160*
	(0.0067)	(0.0092)	(0.0092)
Surprise × NFA × Post2008		0.0222	0.0017
		(0.0156)	(0.0263)
NFA/GDP	0.0004	-0.0010	-0.0011
	(0.0004)	(0.0007)	(0.0007)
Post2008		0.0955*	0.0832*
		(0.0543)	(0.0494)
Post2008 × Surprise		-1.4477	-1.0319
		(1.8043)	(2.3059)
Post2008 × NFA/GDP		0.0010	0.0012*
		(0.0007)	(0.0007)
Country FE	Yes	Yes	Yes
Date Cluster	Yes	Yes	Yes
Observations	2103	2103	2103
R^2	0.0014	0.0070	0.0103

Notes: I define $e_{i,t}$ as foreign currency units per USD. The dependent variable is the spot return $r_{i,t} = -\Delta \log(e_{i,t})$, so $r > 0$ indicates foreign appreciation (USD depreciation). Post2008 = 1 for dates after September 15, 2008 (Lehman collapse). NFA/GDP is demeaned; all specifications include country fixed effects. Standard errors clustered by FOMC date (preferred: common shock structure). *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

8 Table 5: VIX Stress Interaction

Table 8: VIX Stress Interaction in the Balance-Sheet Channel

	(1) Baseline (Date Cl.)	(2) STMT + VIX (Date Cl.)	(3) MP1 + VIX (Date Cl.)
STMT	-0.5978	0.3647	
	(0.6176)	(0.7977)	
MP1			0.2873
			(0.6555)
Surprise × NFA/GDP	-0.0048	0.0013	-0.0136
	(0.0067)	(0.0073)	(0.0093)
Surprise × NFA × HighVIX		-0.0194	-0.0104
		(0.0151)	(0.0231)
NFA/GDP	0.0004	0.0002	0.0001
	(0.0004)	(0.0003)	(0.0003)
HighVIX		0.0881	0.0844
		(0.0762)	(0.0775)
HighVIX × Surprise		-2.0068	-0.7505
		(1.4074)	(0.9623)
HighVIX × NFA/GDP		0.0016	0.0015
		(0.0009)	(0.0009)
Country FE	Yes	Yes	Yes
Date Cluster	Yes	Yes	Yes
Observations	2103	2103	2103
R^2	0.0014	0.0103	0.0133

Notes: I define $e_{i,t}$ as foreign currency units per USD. The dependent variable is the spot return $r_{i,t} = -\Delta \log(e_{i,t})$, so $r > 0$ indicates foreign appreciation (USD depreciation). HighVIX = 1 if VIX $\geq$ 75th percentile of the sample distribution. NFA/GDP is demeaned; all specifications include country fixed effects. Standard errors clustered by FOMC date. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

9 Table 8: Placebo Test

Table 9: Placebo Test: Does Future STMT Predict Current FX Returns?

	Contemporaneous STMT_t	Lead Shock STMT_{t+1}	Both $\text{STMT}_t + \text{STMT}_{t+1}$
STMT_t (Contemp.)	0.490		0.360
	(0.707)		(0.683)
STMT_{t+1} (Lead)		1.759	1.733
		(1.531)	(1.510)
Currency FE	Yes	Yes	Yes
Clustered SE (Date)	Yes	Yes	Yes
Observations	1824	1824	1824
R^2	0.001	0.009	0.009
p-value ($\beta_{t+1} = 0$)		0.251	0.251
Wald p-value ($\beta_t = \beta_{t+1}$)			0.334

Notes: Panel regression of FX returns (in bps) on STMT surprise (in bps). The lead coefficient is imprecisely estimated (a noisy zero), consistent with sampling noise in daily FX. The Wald test fails to reject equality of coefficients, providing no evidence that future shocks have greater predictive power. Standard errors clustered by FOMC date.

Part II

Figures

10 Figure 1: Surprise Time Series

Figure 1: Monetary policy surprise time series (STMT, MP1, MP2), 1994–2024.

11 Figure 2: Correlation Heatmap

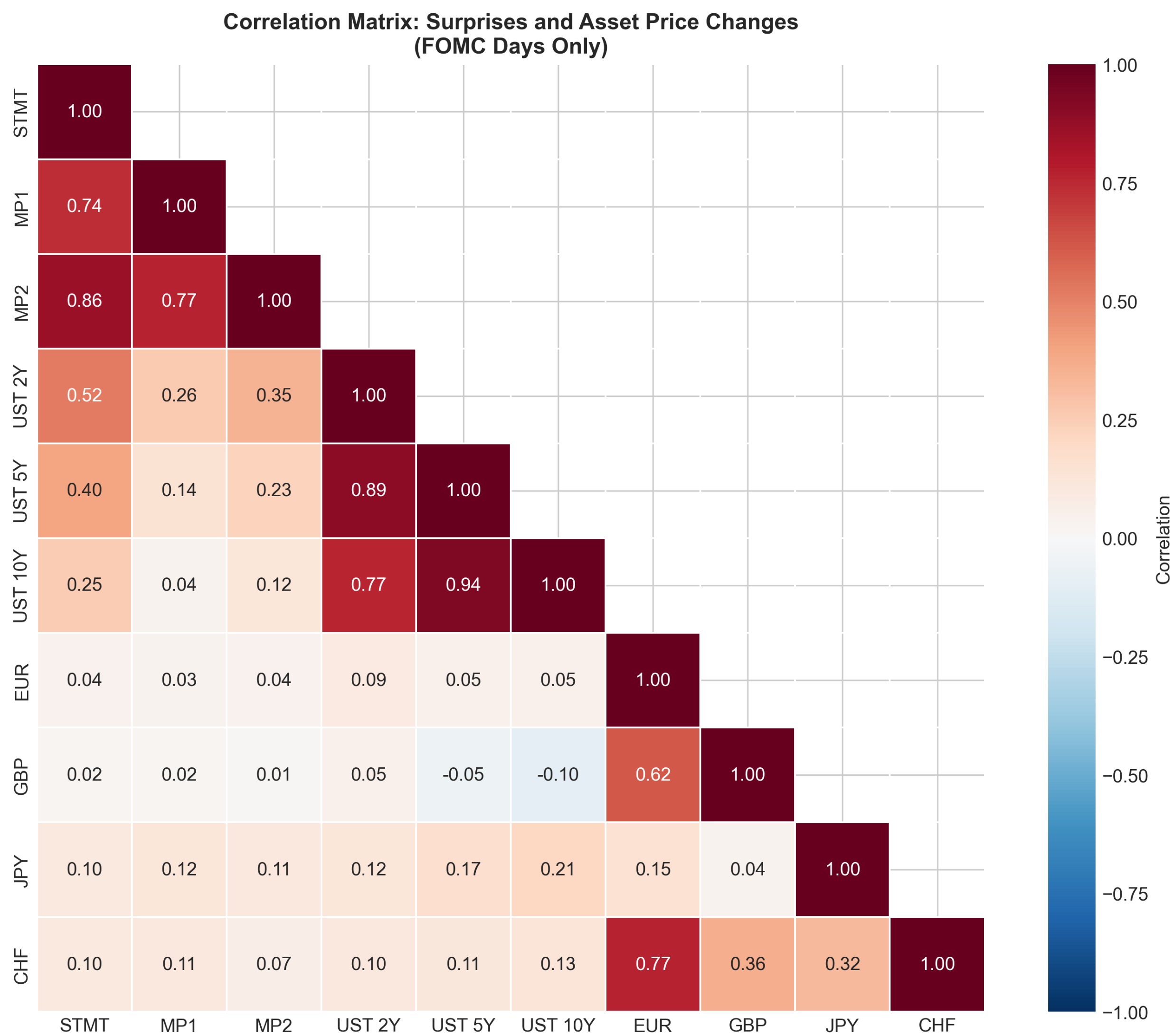

Figure 2: Correlation heatmap of monetary policy surprises and asset price changes.

12 Figure 3: Surprise Distributions

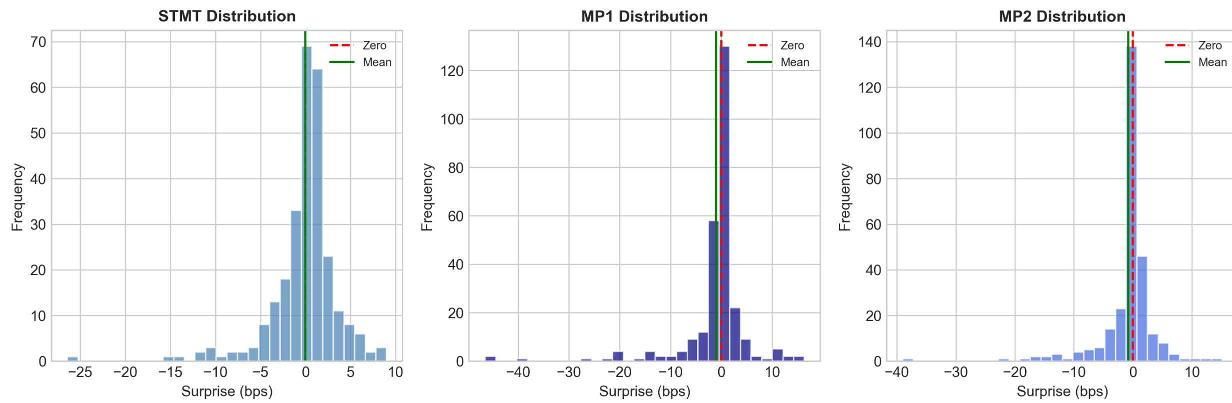

Figure 3: Distribution of monetary policy surprise measures.

13 Figure 4: STMT Validation

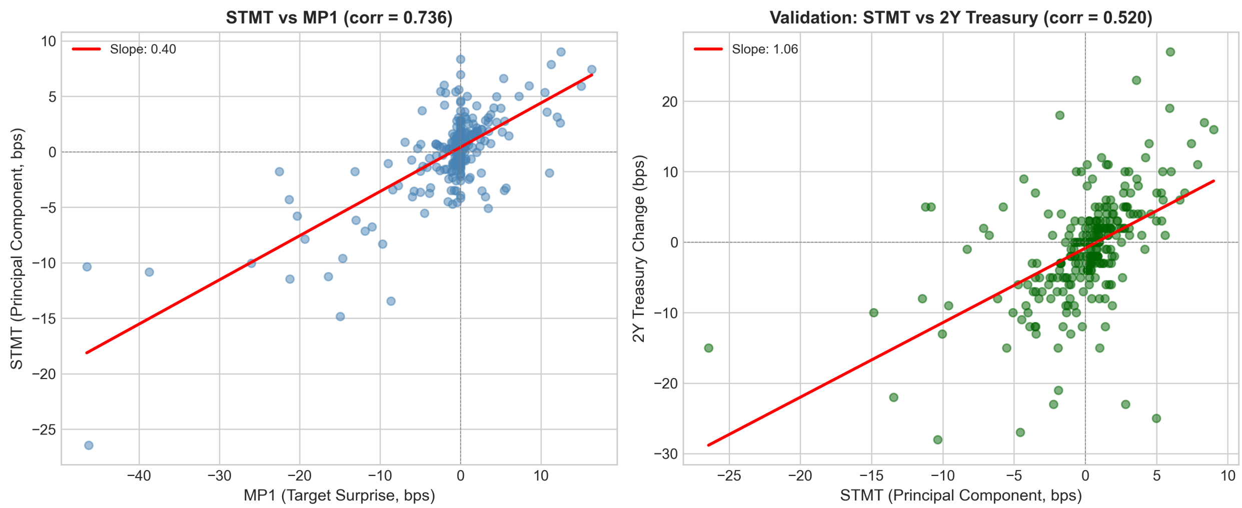

Figure 4: STMT validation: 2-year Treasury response to monetary policy surprises.

14 Figure 5: Event Scatter

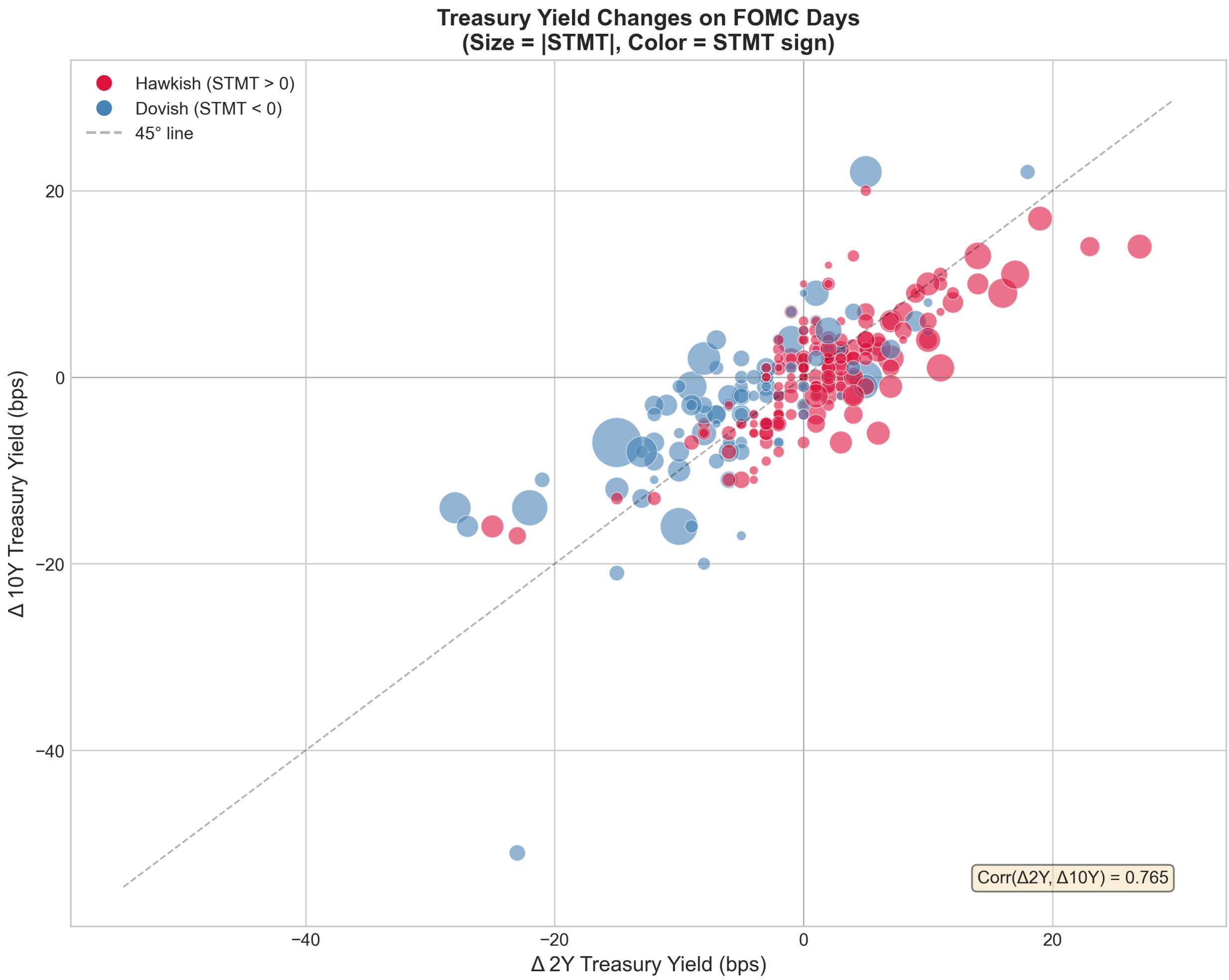

Figure 5: Scatter plot of FX changes vs. monetary policy surprises on FOMC dates.

15 Figure 6: Volatility by Period

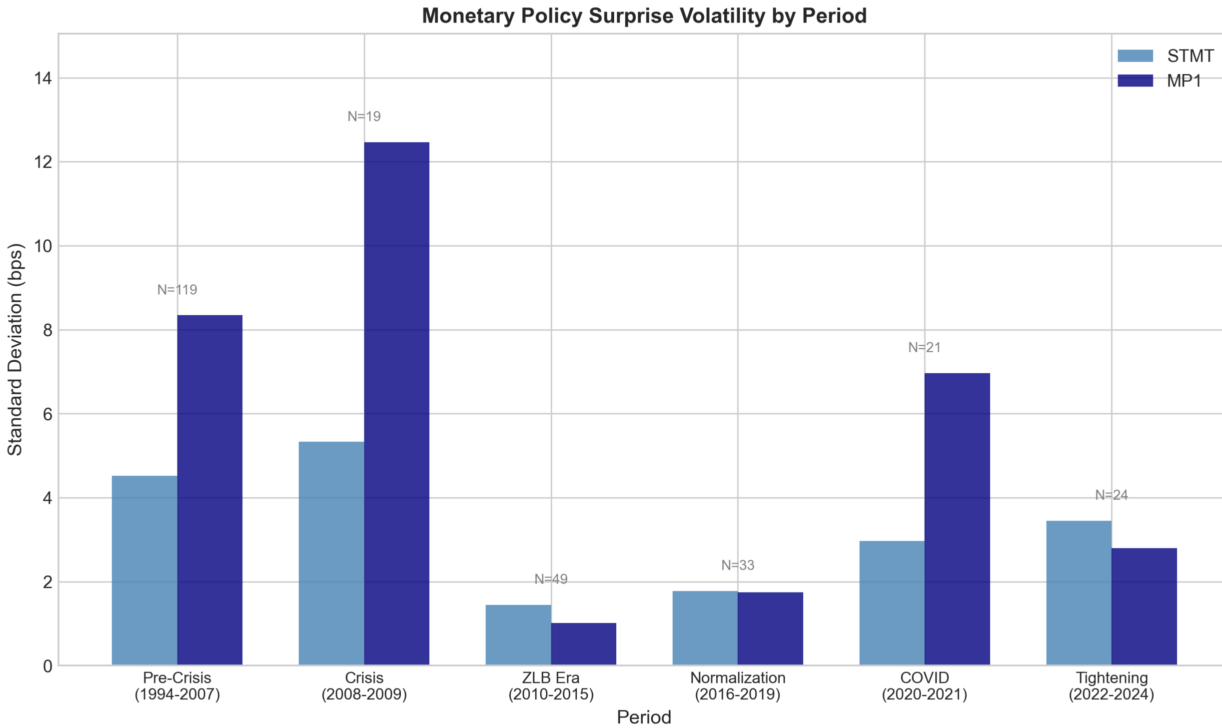

Figure 6: FX volatility by sub-period (pre-GFC, GFC, post-GFC).

16 Figure 7: Term Structure Response

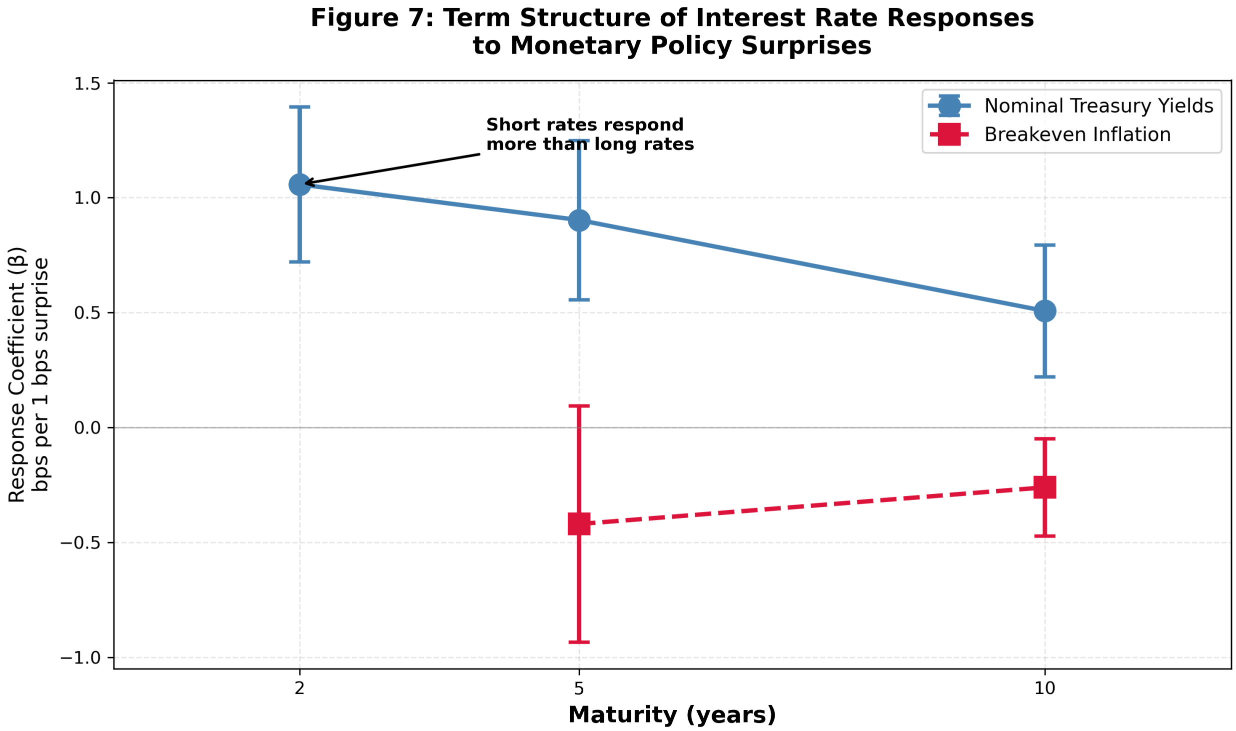

Figure 7: Treasury yield response across the term structure.

17 Figure 8: FX Heterogeneity

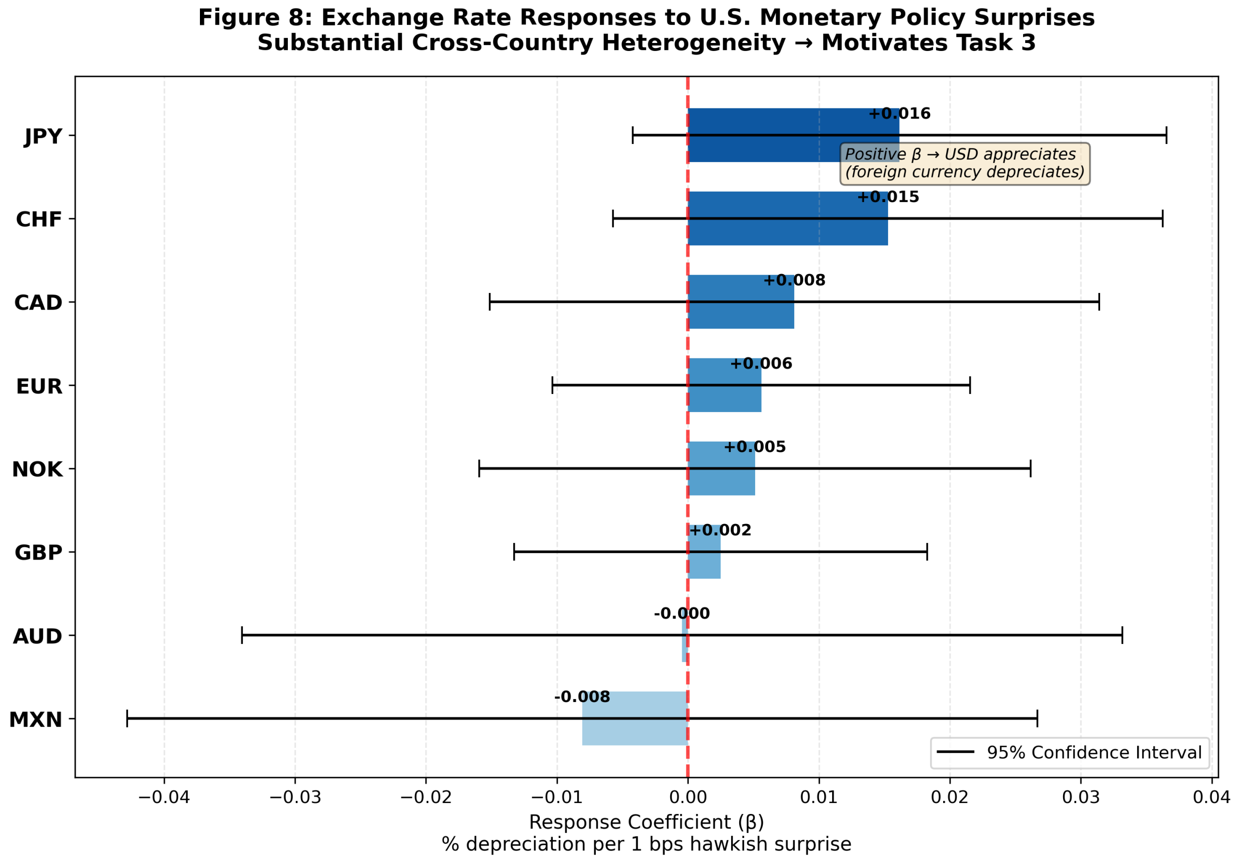

Figure 8: Cross-currency heterogeneity in FX response to monetary policy surprises.

18 Figure 9: Coefficient Plot

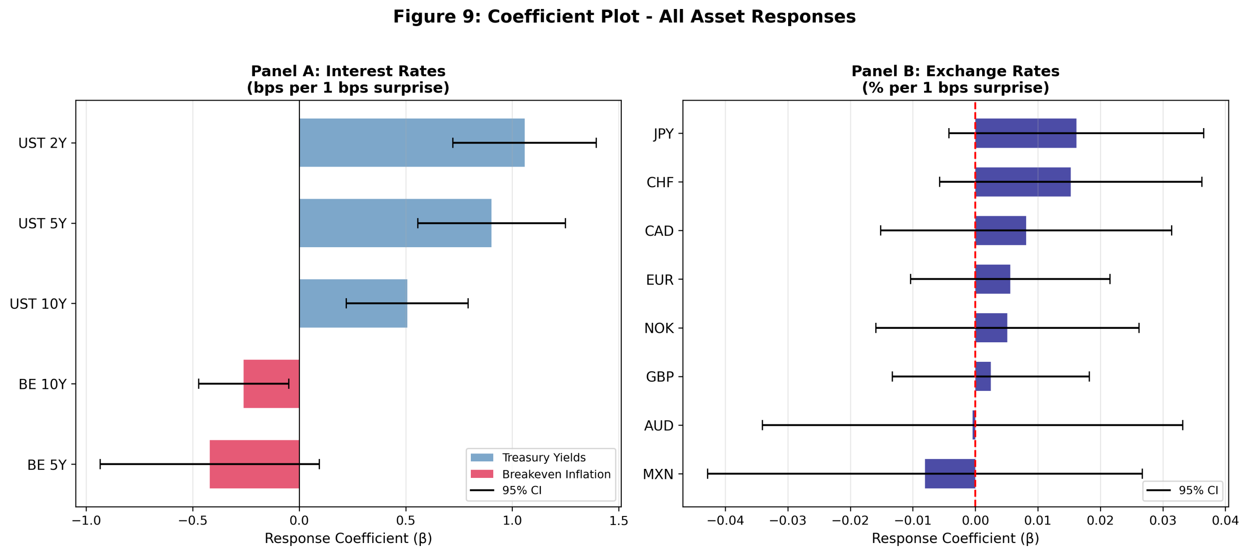

Figure 9: Regression coefficient estimates with 95% confidence intervals.

19 Figure 10: Betas vs NFA

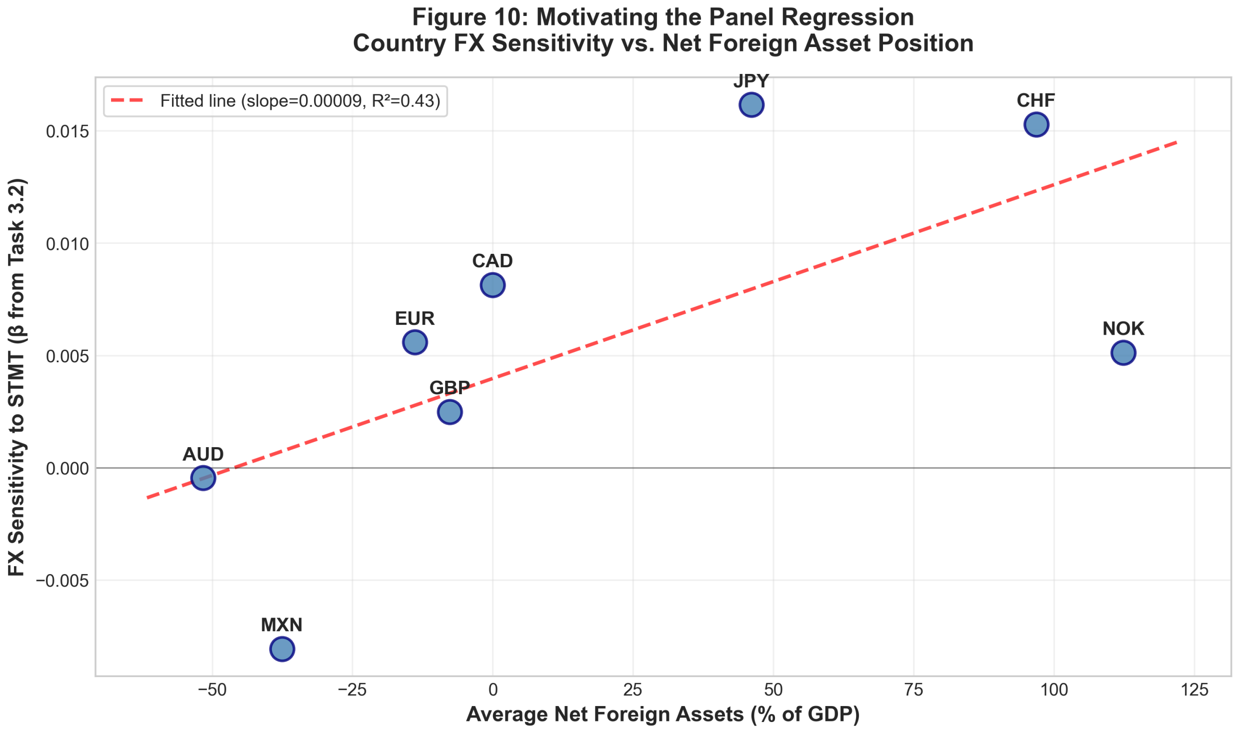

Figure 10: Country-specific FX betas vs. NFA/GDP position.

20 Figure 10b: Betas vs NFA (Both Measures)

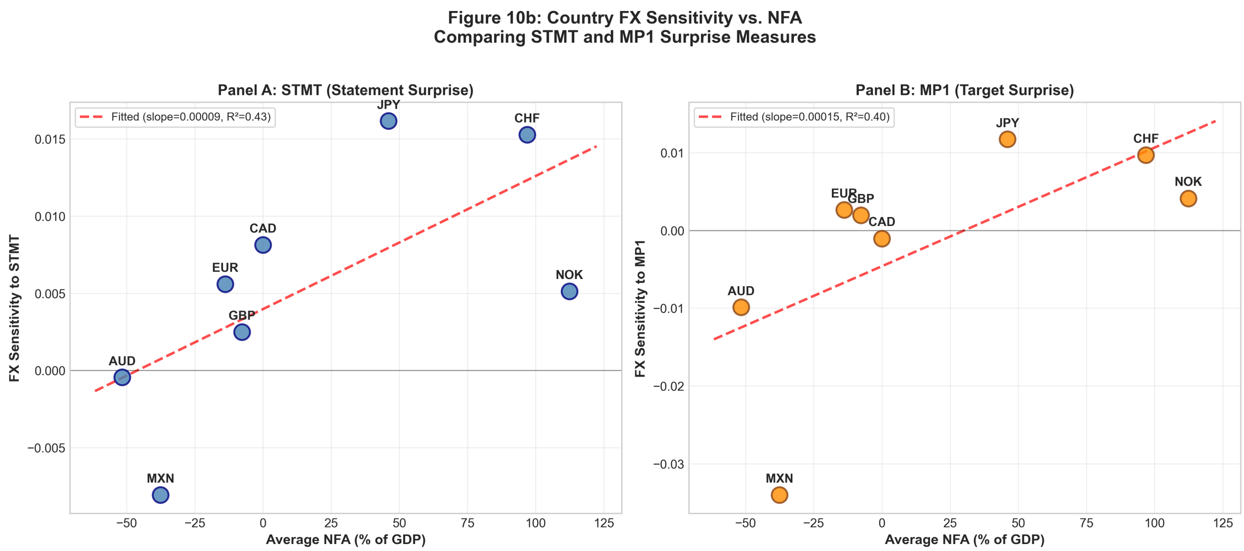

Figure 11: Country-specific FX betas vs. NFA/GDP: STMT and MP1 comparison.

21 Figure 11: Marginal Effects

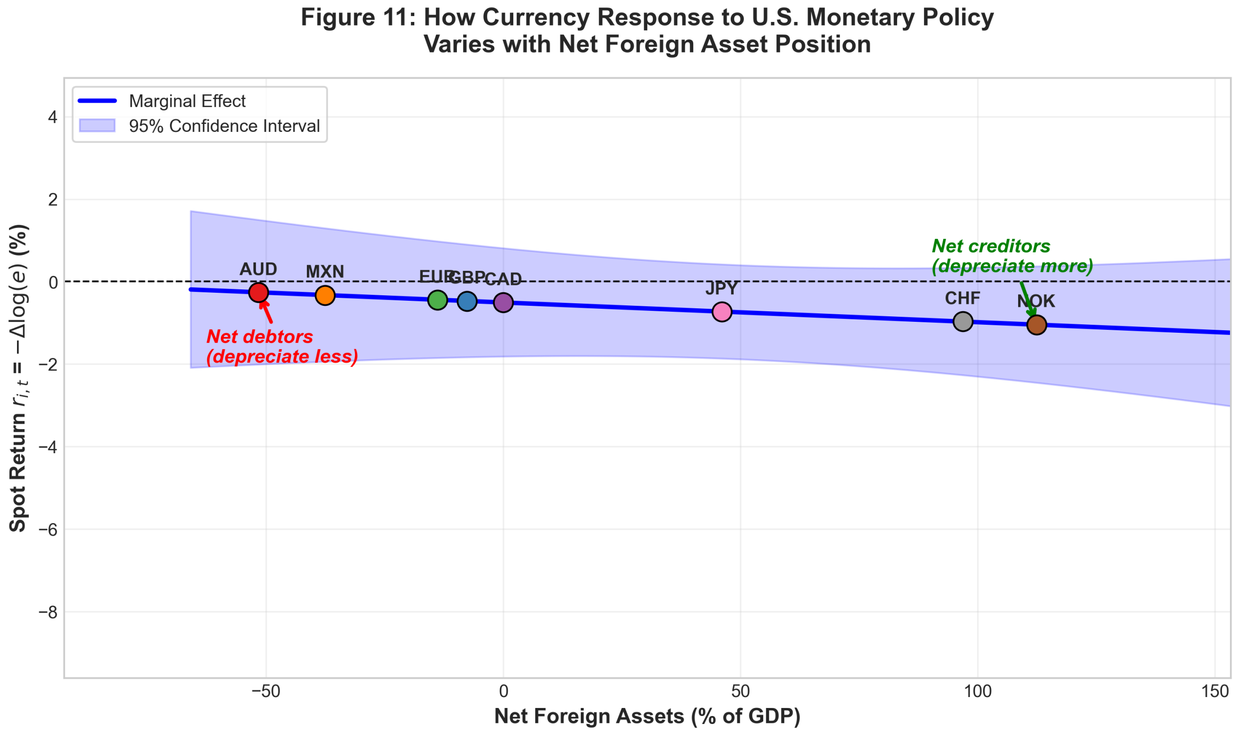

Figure 12: Marginal effects of monetary policy surprise conditional on NFA/GDP.

22 Figure 12: Predicted Responses

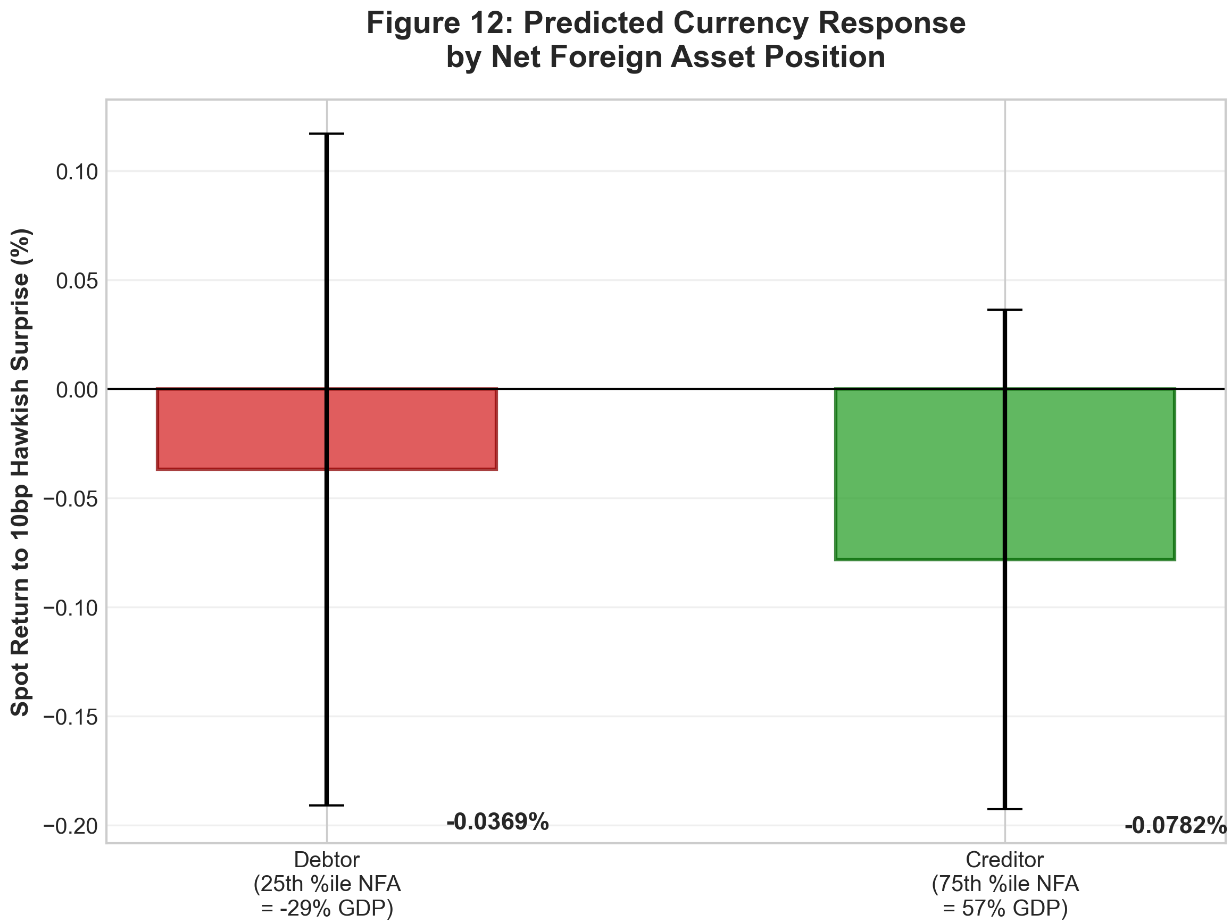

Figure 13: Model-predicted FX responses by NFA quintile.

23 Figure 13: Time Variation

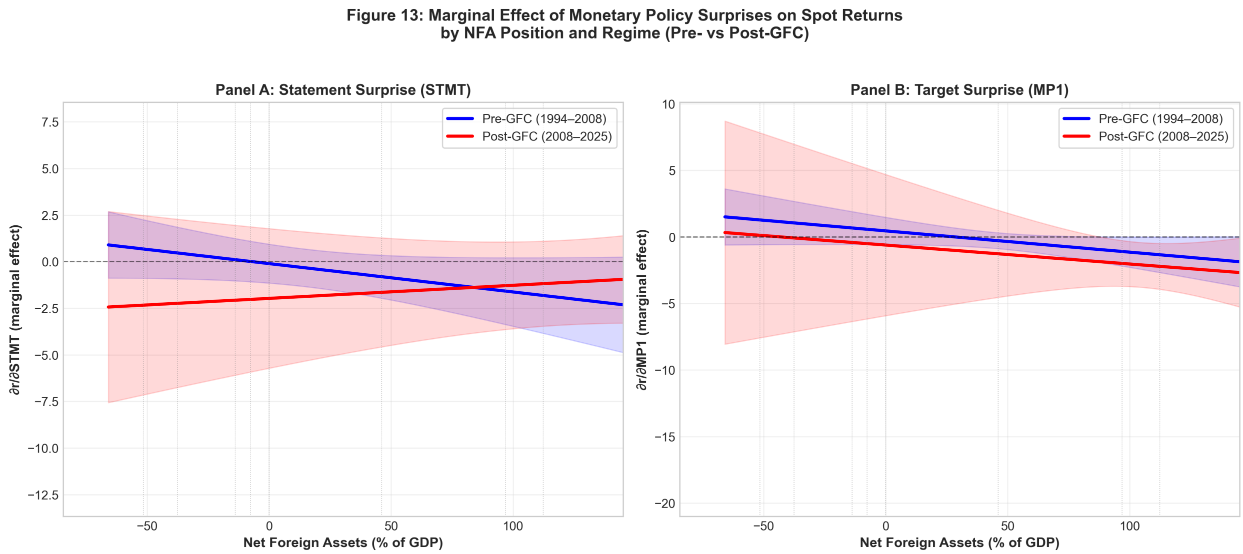

Figure 14: Time variation in NFA channel: pre-GFC vs. post-GFC comparison.

24 Figure 14: Coefficient Comparison

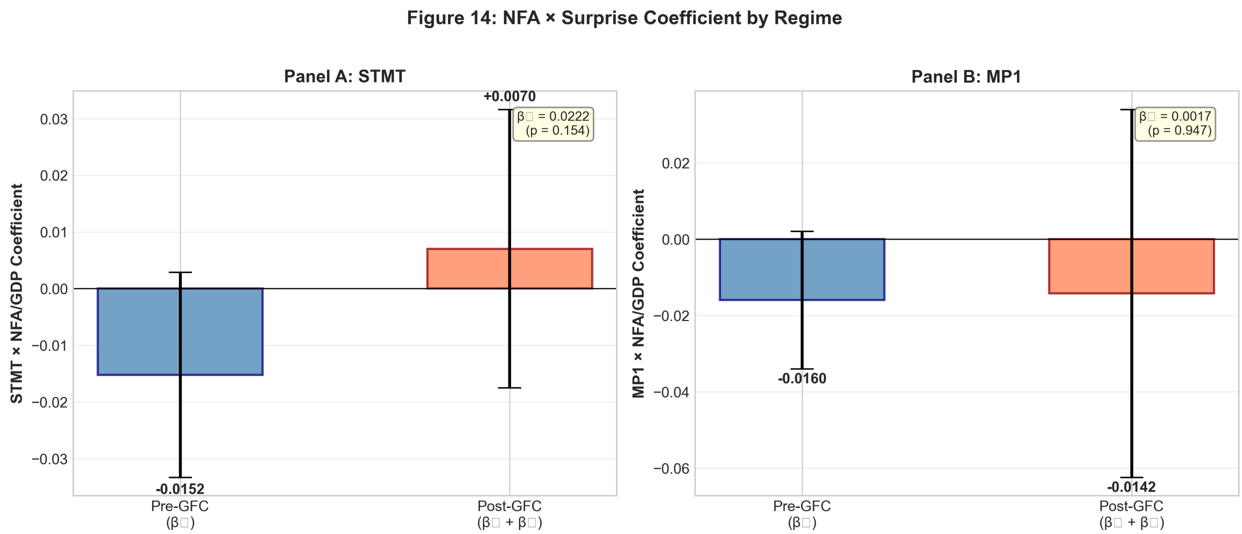

Figure 15: Coefficient comparison across model specifications.

25 Figure 15: VIX Stress

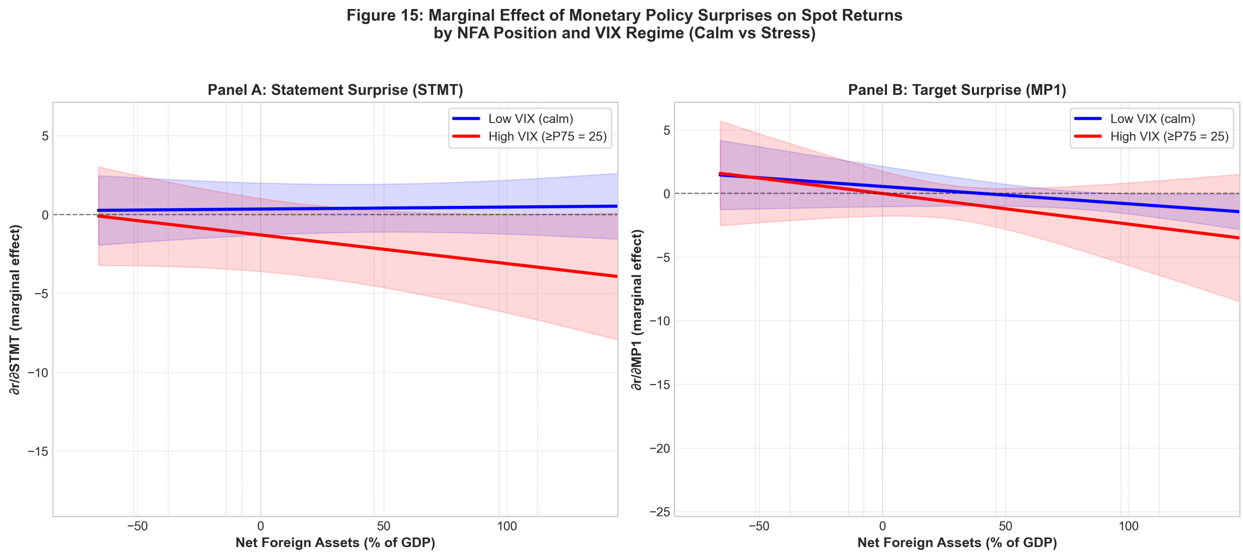

Figure 16: NFA channel during high-VIX stress episodes.

26 Figure 16: VIX Coefficient Comparison

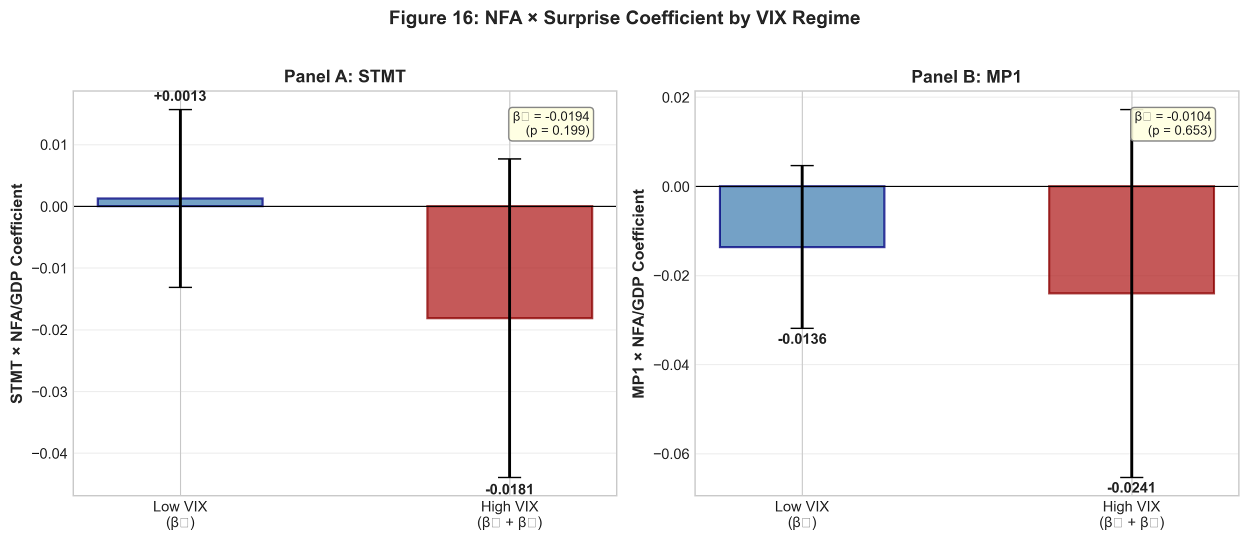

Figure 17: Coefficient comparison: normal vs. high-VIX regimes.

Part III
Supplementary Materials

27 Writeup

The complete analysis writeup is provided as a separate document:

- **writeup.pdf** — Main submission (∼5 pages text + tables/figures)

28 Code Repository

All source code is available at: `https://github.com/Toba4366/MIT-Coding-Challenge`

29 AI Assistance

AI assistance was used in this project. Conversation logs are available at `https://chatgpt.com/share/69980c6a-51cc-8012-b12c-9eac1e7f4b44` and in the References/AI folder.